

Mario Irigoyen Estrada

Actividad 5.4 Instalación de la plataforma de código abierto Docker.

En archlinux el manejador de paquetes lo instala todo con un solo comando

```
~
> sudo pacman -S docker
[sudo] password for tarmairon:
resolving dependencies ...
looking for conflicting packages ...

Packages (3) containerd-2.2.3-1 runc-1.4.2-1 docker-1:29.4.0-1

Total Installed Size: 215.14 MiB

:: Proceed with installation? [Y/n] y
(3/3) checking keys in keyring [#####] 100%
(3/3) checking package integrity [#####] 100%
(3/3) loading package files [#####] 100%
(3/3) checking for file conflicts [#####] 100%
(3/3) checking available disk space [#####] 100%
:: Processing package changes ...
(1/3) installing runc [#####] 100%
Optional dependencies for runc
  criu: checkpoint support
(2/3) installing containerd [#####] 100%
(3/3) installing docker [#####] 100%
Optional dependencies for docker
  btrfs-progs: btrfs backend support
  pigz: parallel gzip compressor support
  docker-buildx: extended build capabilities [installed]
:: Running post-transaction hooks ...
(1/3) Creating system user accounts...
(2/3) Reloading system manager configuration...
(3/3) Arming ConditionNeedsUpdate...
```

Pero docker compose es un paquete aparte

```
~ took 5s
> sudo pacman -S docker-compose
warning: docker-compose-5.1.3-1 is up to date -- reinstalling
resolving dependencies ...
looking for conflicting packages ...

Packages (1) docker-compose-5.1.3-1

Total Installed Size: 28.21 MiB
Net Upgrade Size: 0.00 MiB

:: Proceed with installation? [Y/n] y
(1/1) checking keys in keyring [#####] 100%
(1/1) checking package integrity [#####] 100%
(1/1) loading package files [#####] 100%
(1/1) checking for file conflicts [#####] 100%
(1/1) checking available disk space [#####] 100%
:: Processing package changes ...
(1/1) reinstalling docker-compose [#####] 100%
:: Running post-transaction hooks ...
(1/1) Arming ConditionNeedsUpdate...
```

```
~
> docker --version
Docker version 29.4.0, build 9d7ad9ff18

~
> docker compose version
Docker Compose version 5.1.3

~
```

Preguntas

Como se pueden ver los contenedores que tenemos creados sin docker desktop?

Se pueden ver con el comando **docker ps** para ver los contenedores que estan corriendo activamente y el comando **docker ps -a** para ver todos los contenedores creados

```
~
> docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
708637f2b2d7   ds-frontend-app                    "/docker-entrypoint..." 4 months ago   Up 14 hours   0.0.0.0:4200→80/tcp, [::]:4200→80/t
cp             ds_frontend
6b89dd85e568   ds-backend-app                    "./main"                 4 months ago   Up 14 hours   0.0.0.0:3001→8080/tcp, [::]:3001→80
80/tcp        ds_backend
2a9fd79a0340   mysql:8.0                          "docker-entrypoint.s..." 4 months ago   Up 14 hours (healthy) 33060/tcp, 0.0.0.0:3308→3306/tcp, [::]:3308→3306/tcp
ds_database

~
> docker ps -a
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
d0c1ffdc1986   ubuntu                             "/bin/bash"             3 months ago   Exited (0) 3 months ago
9178391c64c7   ubuntu                             "-it /bin/bash"         3 months ago   Created
beautiful_maxwell
708637f2b2d7   ds-frontend-app                    "/docker-entrypoint..." 4 months ago   Up 14 hours   0.0.0.0:4200→80/tcp, [::]:4200→80
/tcp        ds_frontend
6b89dd85e568   ds-backend-app                    "./main"                 4 months ago   Up 14 hours   0.0.0.0:3001→8080/tcp, [::]:3001→
8080/tcp    ds_backend
2a9fd79a0340   mysql:8.0                          "docker-entrypoint.s..." 4 months ago   Up 14 hours (healthy) 33060/tcp, 0.0.0.0:3308→3306/tcp, [::]:3308→3306/tcp
ds_database
```

Como se pueden ver las imagenes que tenemos en nuestro local?

Se pueden ver con el comando **docker images**, si queremos remover una imagen se usa el comando **docker rmi <nombre de la imagen/sha>**

```
> docker images
IMAGE                                ID                                DISK USAGE  CONTENT SIZE  In Use
agenda-app:latest                   4e9c6c546e19                     675MB       0B
agenda-db:latest                     4fb41fc48655                     783MB       0B
agenda-nginx:local                   f18969392ac8                     57MB        0B
agenda-php:local                     93cc195557ee                     675MB       0B
agenda-web:latest                   599146411557                     56.9MB      0B
agenda:latest                       c24b39703f1b                     56.9MB      0B
alpine:latest                       706db57fb206                     8.32MB      0B
docker:24.0.5                       7015f2c475d5                     335MB       0B
docker:24.0.5-dind                  7015f2c475d5                     335MB       0B
ds-backend-app:latest               aa43aa80935d                     39.5MB      0B
ds-backend-backend:latest           1b2718d77d05                     37.7MB      0B
ds-frontend-app:latest              14da4f5d31d2                     63.6MB      0B
gitlab/gitlab-runner:alpine         ce9bb70ca38f                     194MB       0B
mysql:8.0                           34178dbaefd0                     783MB       0B
nginx:alpine                        5e7abdd2021                      52.7MB      0B
php:8.3-cli                         c0c73a8292a4                     532MB       0B
php:8.4-fpm                         81907c64e447                     498MB       0B
registry.gitlab.com/gitlab-org/gitlab-runner/gitlab-runner-helper:x86_64-v18.4.0 36d8f9ccf058                     90.4MB      0B
registry.gitlab.com/kokkoann/agenda/mysql:latest 551b98b7da0f                     783MB       0B
registry.gitlab.com/kokkoann/agenda/nginx:main-3876189f bdcffa201f68                     55.6MB      0B
registry.gitlab.com/kokkoann/agenda/php:main-3876189f 11fb00c2ceae                     666MB       0B
ubuntu:latest                       c3a134f2ace4                     78.1MB      0B
```

Como se se pueden ver los logs de los contenedores?

El comando **docker logs <nombre del contenedor/sha>** muestra los logs de la aplicacion que van directo a stdout, igual se puede agregar la bandera **-f** para verlos en tiempo real, si necesitamos ver un archivo de logs dentro del contenedor se tiene que ejecutar el comando **docker exec <nombre del contenedor/sha> cat <archivo>**, en

caso de querer usar cat. Igual se puede entrar al contenedor a alguna shell, los de alpine no suelen tener bash, con el comando **docker exec -it <nombre del contenedor/sha> /bin/sh**

```
> docker logs -n 15 ds_backend
[GIN-debug] PUT    /api/v1/contratos/:id    → backend/internal/controllers.(*ContratosController).UpdateContrato-fm (6 handlers)
[GIN-debug] DELETE /api/v1/contratos/:id    → backend/internal/controllers.(*ContratosController).DeleteContrato-fm (6 handlers)
[GIN-debug] GET    /api/v1/imagenes/all/propiedad/:id → backend/internal/controllers.(*ImagenesController).GetImagenesByPropiedad-fm (6 handlers)
[GIN-debug] GET    /api/v1/imagenes/all/principal/:id → backend/internal/controllers.(*ImagenesController).GetImagenPrincipal-fm (6 handlers)
[GIN-debug] POST   /api/v1/imagenes/create  → backend/internal/controllers.(*ImagenesController).InsertImagen-fm (6 handlers)
[GIN-debug] DELETE /api/v1/imagenes/eliminar/:id → backend/internal/controllers.(*ImagenesController).DeleteImagen-fm (6 handlers)
[GIN-debug] GET    /api/v1/documentos_anexos/all/propiedad/:id → backend/internal/controllers.(*DocumentosAnexosController).GetDocumentosByPropiedad-fm (6 handlers)
[GIN-debug] GET    /api/v1/documentos_anexos/:id → backend/internal/controllers.(*DocumentosAnexosController).GetDocumentoAnexo-fm (6 handlers)
[GIN-debug] POST   /api/v1/documentos_anexos/create → backend/internal/controllers.(*DocumentosAnexosController).InsertDocumentoAnexo-fm (6 handlers)
[GIN-debug] POST   /api/v1/email/enviar-email-verificacion → backend/internal/controllers.(*VerificarEmailController).EnviarEmailVerificacion-fm (6 handlers)
[GIN-debug] POST   /api/v1/email/verificar-email → backend/internal/controllers.(*VerificarEmailController).VerificarEmail-fm (6 handlers)
[GIN-debug] POST   /api/v1/email/reenviar-codigo-verificacion → backend/internal/controllers.(*VerificarEmailController).ReenviarCodigoVerificacion-fm (6 handlers)
[GIN-debug] [WARNING] You trusted all proxies, this is NOT safe. We recommend you to set a value.
Please check https://pkg.go.dev/github.com/gin-gonic/gin#readme-don-t-trust-all-proxies for details.
[GIN-debug] Listening and serving HTTP on :8080
```

Como se pueden pasar imagenes como archivos?

Para generar un archivo de una imagen, ya sea como backup o si no se puede utilizar un repositorio en linea, se utiliza el comando **docker image save <imagen> > <archivo>.tar** para guardar la imagen en el local se utiliza la imagen **docker image load <archivo>.tar**

<https://docs.docker.com/reference/cli/docker/image/save/>

<https://docs.docker.com/reference/cli/docker/image/load/>

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```

> docker images

IMAGE                                         ID                                         DISK USAGE   CONTENT SIZE  EXTRA
agenda-app:latest                          4e9c6c546e19                             675MB        0B
agenda-db:latest                           4fb41fc48655                             783MB        0B
agenda-nginx:local                         f18969392ac8                             57MB         0B
agenda-php:local                           93cc195557ee                             675MB        0B
agenda-web:latest                          599146411557                             56.9MB       0B
agenda:latest                              c24b39703f1b                             56.9MB       0B
alpine:latest                             706db57fb206                             8.32MB       0B
docker:24.0.5                              7015f2c475d5                             335MB        0B
docker:24.0.5-dind                        7015f2c475d5                             335MB        0B
ds-backend-app:latest                     aa43aa80935d                             39.5MB       0B
ds-backend-backend:latest                 1b2718d77d05                             37.7MB       0B
ds-frontend-app:latest                    14da4f5d31d2                             63.6MB       0B
gitlab/gitlab-runner:alpine               ce9bb70ca38f                             194MB        0B
mysql:8.0                                 34178dbaefd0                             783MB        0B
nginx:alpine                              5e7abccd2021                             52.7MB       0B
php:8.3-cli                              c0c73a8292a4                             532MB        0B
php:8.4-fpm                              81907c64e447                             498MB        0B
registry.gitlab.com/gitlab-org/gitlab-runner/gitlab-runner-helper:x86_64-v18.4.0 36d8f9ccf058                             90.4MB       0B
registry.gitlab.com/kokkoann/agenda/mysql:latest 551b98b7da0f                             783MB        0B
registry.gitlab.com/kokkoann/agenda/nginx:main-3876189f bdcffa201f68                             55.6MB       0B
registry.gitlab.com/kokkoann/agenda/php:main-3876189f 11fb00c2ceae                             666MB        0B
ubuntu:latest                             c3a134f2ace4                             78.1MB       0B

~
> docker image save php:8.4-fpm > php-image.tar

~
> ls -S php-image.tar
-rw-rw-r-- 1 php-image.tar

~
> docker rmi php:8.4-fpm
Untagged: php:8.4-fpm
Untagged: php@sha256:231ab3ac3db542e1d019a35abee1bd106d278a5712d1dc2bb4fbe103de016792
Deleted: sha256:81907c64e447160ece0de319c62ab29b50f1c4b6b9db6ee1ed6c78b9b868fb4c

~
> docker images

IMAGE                                         ID                                         DISK USAGE   CONTENT SIZE  EXTRA
agenda-app:latest                          4e9c6c546e19                             675MB        0B
agenda-db:latest                           4fb41fc48655                             783MB        0B
agenda-nginx:local                         f18969392ac8                             57MB         0B
agenda-php:local                           93cc195557ee                             675MB        0B
agenda-web:latest                          599146411557                             56.9MB       0B
agenda:latest                              c24b39703f1b                             56.9MB       0B
alpine:latest                             706db57fb206                             8.32MB       0B
docker:24.0.5                              7015f2c475d5                             335MB        0B
docker:24.0.5-dind                        7015f2c475d5                             335MB        0B
ds-backend-app:latest                     aa43aa80935d                             39.5MB       0B
ds-backend-backend:latest                 1b2718d77d05                             37.7MB       0B
ds-frontend-app:latest                    14da4f5d31d2                             63.6MB       0B
gitlab/gitlab-runner:alpine               ce9bb70ca38f                             194MB        0B
mysql:8.0                                 34178dbaefd0                             783MB        0B
nginx:alpine                              5e7abccd2021                             52.7MB       0B
php:8.3-cli                              c0c73a8292a4                             532MB        0B
registry.gitlab.com/gitlab-org/gitlab-runner/gitlab-runner-helper:x86_64-v18.4.0 36d8f9ccf058                             90.4MB       0B
registry.gitlab.com/kokkoann/agenda/mysql:latest 551b98b7da0f                             783MB        0B
registry.gitlab.com/kokkoann/agenda/nginx:main-3876189f bdcffa201f68                             55.6MB       0B
registry.gitlab.com/kokkoann/agenda/php:main-3876189f 11fb00c2ceae                             666MB        0B
ubuntu:latest                             c3a134f2ace4                             78.1MB       0B

~

```

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```
> docker load -i php-image.tar
Loaded image: php:8.4-fpm

> docker images
```

IMAGE	ID	DISK USAGE	Info	In Use
			CONTENT SIZE	EXTRA
agenda-app:latest	4e9c6c546e19	675MB	0B	
agenda-db:latest	4fb41fc48655	783MB	0B	
agenda-nginx:local	f18969392ac8	57MB	0B	
agenda-php:local	93cc195557ee	675MB	0B	
agenda-web:latest	599146411557	56.9MB	0B	
agenda:latest	c24b39703f1b	56.9MB	0B	
alpine:latest	706db57fb206	8.32MB	0B	
docker:24.0.5	7015f2c475d5	335MB	0B	
docker:24.0.5-dind	7015f2c475d5	335MB	0B	
ds-backend-app:latest	aa43aa80935d	39.5MB	0B	U
ds-backend-backend:latest	1b2718d77d05	37.7MB	0B	
ds-frontend-app:latest	14da4f5d31d2	63.6MB	0B	U
gitlab/gitlab-runner:alpine	ce9bb70ca38f	194MB	0B	
mysql:8.0	34178dbaefd0	783MB	0B	U
nginx:alpine	5e7abcedd2021	52.7MB	0B	
php:8.3-cli	c0c73a8292a4	532MB	0B	
php:8.4-fpm	81907c64e447	498MB	0B	
registry.gitlab.com/gitlab-org/gitlab-runner/gitlab-runner-helper:x86_64-v18.4.0	36d8f9ccf058	90.4MB	0B	
registry.gitlab.com/kokkoann/agenda/mysql:latest	551b98b7da0f	783MB	0B	
registry.gitlab.com/kokkoann/agenda/nginx:main-3876189f	bdcffa201f68	55.6MB	0B	
registry.gitlab.com/kokkoann/agenda/php:main-3876189f	11fb00c2ceae	666MB	0B	
ubuntu:latest	c3a134f2ace4	78.1MB	0B	U